SOLAR PROPOSAL DECODER

A Homeowner's System for Comparing Quotes, Financing, Production Claims, Scope, and Warranties Before Signing

FIND	EXTRACT	NORMALIZE	CHALLENGE
ASK	VERIFY	DOCUMENT	DECIDE

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Independent consumer study aid

READER NOTICE

This book is an independent educational tool for U.S. homeowners comparing residential rooftop-solar proposals. It is not affiliated with or endorsed by the U.S. Department of Energy, the Consumer Financial Protection Bureau, the Federal Trade Commission, the National Renewable Energy Laboratory, any utility, lender, equipment manufacturer, or installer.

The book helps you locate, copy, compare, and question information supplied by sellers and lenders. It does not select an installer, recommend a loan, determine tax-credit eligibility, calculate your personal return, certify a roof or electrical design, predict utility savings, interpret a contract for you, or replace advice from qualified legal, tax, financial, engineering, roofing, insurance, or electrical professionals.

Programs, incentives, utility rules, financing products, and company terms change. Verify them with the responsible authority or provider before relying on them. In every transaction, the signed installation contract and the lender's or third-party owner's documents control—not a sales slide, verbal statement, or this book.

All proposal fragments, people, companies, prices, equipment names, addresses, and scenarios in this book are synthetic. They are designed to teach a method and do not describe a real offer.

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How this book earns its place next to your proposals

A spreadsheet can compare numbers after you know which numbers belong together. A calculator can estimate production after you choose assumptions. A marketplace can make offers from participating sellers look more uniform. None of those tools necessarily teaches you how to take three differently structured proposals, locate their controlling facts, expose hidden assumptions, and turn each unresolved gap into a written question.

That is the job of this book.

The workflow is deliberately repetitive at the level where repetition helps:

REAL PROPOSAL → FIND → EXTRACT → NORMALIZE → CHALLENGE → WRITTEN QUESTIONS → ANSWERS OR REVISED PROPOSALS → DECISION PACKET

You will first learn the method. Then you will apply it to a synthetic three-quote case. Finally, you will use a write-in packet with your own proposals. The reference pages come before the forms because a blank box is useful only when you know what belongs in it.

Use pencil for your first pass. Write proposal page numbers beside every copied value. Keep screenshots or PDFs of portal-only material. Date every revised offer. If an answer changes the price, scope, equipment, production estimate, payment schedule, or warranty, ask for a revised document rather than relying on an email sentence alone.

> THE PAUSE RULE
> If a material price, payment, ownership, production, scope, roof, electrical, warranty, cancellation, transfer, or incentive question is still unanswered, the comparison is not finished. An attractive headline is not a reason to rush an incomplete decision.

PART I — ORIENT

1. The decision is not one number

Solar proposals invite fast comparison. One seller leads with a monthly payment. Another leads with annual production. A third displays a net price after an assumed incentive. Each headline may be mathematically defensible inside its own assumptions, yet the three headlines do not answer the same question.

The practical task is to create one common evidence set. For each proposal you need to identify at least six layers:

- commercial structure: cash purchase, loan, lease, power purchase agreement, or another arrangement;
- system definition: DC size, major equipment, quantities, and design basis;
- production model: estimated output and the assumptions behind it;
- physical and administrative scope: what work is included, excluded, allowed for, or subject to change;
- warranties and responsibility: what is covered, by whom, for how long, with which remedy and exclusions;
- transaction terms: payment schedule, cancellation, project changes, transfer, sale of the home, and the controlling documents.

These layers interact. A lower price can exclude work another quote includes. A lower monthly payment can come from a longer term or a payment change. A higher production estimate can come from a stronger design, a different weather file, a different shade model, or a more optimistic loss assumption. A 25-year headline can describe panel performance while installer workmanship lasts far less time.

The system in this book does not collapse all of that into a magic score. It makes the differences visible so that you can decide which differences matter to you.

The four evidence labels

Mark every important item with one of four labels:

- FACT — a defined value or obligation stated in a document, such as 8.2 kW DC or a quoted cash price;
- ESTIMATE — a modeled or forecast value, such as Year-1 production or projected utility escalation;
- CLAIM — persuasive language that requires a definition or supporting term, such as “turnkey” or “100% offset”;
- MISSING — a material field you cannot locate or calculate from the documents supplied.

A fact can still be incomplete. A contract may state a price without explaining change-order exposure. An estimate can be useful without being guaranteed. A claim is not automatically false. Missing information is not proof of an unfavorable answer; it is a reason to ask.

The source-reference habit

Never copy a value without noting where it came from. Use a short reference such as "Proposal A, p. 6, Cash Option" or "Loan disclosure, p. 2." This prevents a common failure: remembering the salesperson's explanation while losing the document that supposedly supports it.

When two documents conflict, do not choose the more favorable one. Record the conflict and ask which term will control. If a proposal says "fixed payment" and the lender schedule shows two payment levels, the discrepancy is itself a decision fact.

> QUICK TEST
> Can another adult pick up your packet and find the source for the cash price, financed principal, production estimate, major exclusions, and warranty terms in under five minutes? If not, your extraction is not yet auditable.

2. Choose the right comparison lane

Do not force unlike commercial structures into one shallow grid. A cash purchase and a loan both generally lead toward ownership, but their cash-flow and credit costs differ. A lease generally means paying to use a system owned by another party. Under a power purchase agreement, the customer generally buys the electricity produced under contract terms. Ownership, maintenance, incentives, escalation, transfer, and end-of-term obligations can differ materially.

Start by writing the transaction type at the top of each proposal. If the type is unclear, that is Question 1.

Lane A: cash purchase

The clean starting figure is the full cash contract price before customer-specific tax assumptions. Then identify deposits, progress payments, allowances, possible change orders, and excluded work. A "net price" after a modeled incentive is not the same field as the amount you promise to pay the installer.

Lane B: financed purchase

Keep the installation price and financing terms separate. At minimum, locate the cash price for the same scope, amount financed or principal, APR, term, payment schedule, total of payments or equivalent disclosure, fees, any expected large prepayment, any payment change or re-amortization event, security interest, prepayment terms, and home-sale implications.

The CFPB has documented solar-specific loans in which the financed principal exceeds the cash price because fees or markups are embedded in the transaction. That does not tell you what your proposal contains. It tells you to compare the actual cash price and financed principal and ask for the difference to be explained in writing.

Lane C: lease or power purchase agreement

Do not treat “no money down” as “no obligation.” Record contract length, initial payment or energy rate, any escalator, minimum payment or production terms, maintenance responsibility, insurance responsibilities, purchase options, transfer process, early termination, end-of-term removal, and which party receives incentives or environmental attributes.

The arithmetic is different from an owned system. Cost per installed watt may describe the hardware but may not describe what the customer is purchasing. Compare the actual payment obligation and contract rights instead.

When not to combine lanes

If Proposal A is cash, Proposal B is a loan, and Proposal C is a PPA, create two views:

1. a project view for system, production, roof, electrical, scope, schedule, and service;
2. a transaction view for ownership, payments, escalation, incentives, transfer, termination, and end-of-term obligations.

Do not announce a winner until both views are complete.

PART II — FIND AND EXTRACT

3. A proposal is an index, not the whole deal

A polished proposal often combines system design, sales narrative, savings graphics, price options, equipment summaries, and selected warranty headlines. The proposal may point to—or omit—documents that eventually control the transaction: an installation contract, financing agreement, disclosure package, warranty terms, utility application, or change order.

Think of the proposal as an index of statements to locate and verify.

Synthetic proposal anatomy

The following example is original and intentionally compact.

SYNTHETIC PROPOSAL S-1047 — SUNRIDGE HOME ENERGY

Customer: Jordan Lee. Site: 18 Example Street. Proposal valid through October 14.

SYSTEM SUMMARY

8.2 kW DC; 20 Helio H410 modules; GridLink X8 inverter; roof-mounted; monitoring included.

ESTIMATED PERFORMANCE

Year-1 production: 11,480 kWh. Usage baseline: 11,300 kWh. Headline: “102% offset.” Model references a shade study dated September 12.

PURCHASE OPTIONS

Cash price: $24,600 before customer-specific incentives or tax treatment.

Finance option: $31,980 principal; 4.99% stated APR; 20-year term; $211 estimated monthly payment. Separate lender disclosures apply.

INCLUDED WORK

Design; permit filing; standard roof mounting; interconnection submission; monitoring setup; inspection coordination.

WARRANTY SUMMARY

25-year panel performance; 12-year panel product; 10-year installer workmanship; inverter covered under manufacturer terms.

ASSUMPTIONS AND EXCLUSIONS

Main-panel upgrade, roof repair, trenching, tree work, hazardous-material work, and utility-required redesign are excluded. Price may change after site inspection. Customer remains responsible for utility charges not offset by solar production.

This fragment contains useful information. It also creates questions. What exactly changes after site inspection? What is the cash price for the same scope as the financed offer? Does the $211 payment remain fixed? What losses drive the production model? Who pays labor under a manufacturer warranty? How are unexpected roof or electrical conditions handled?

The eight-location sweep

Search every proposal in this order:

1. Cover or executive summary — offer identity, date, customer, address, headline.
2. System page — DC size, module count and model, inverter, battery, layout.
3. Performance page — Year-1 kWh, usage baseline, offset, assumptions, guarantee.
4. Price page — gross cash price, discounts, adders, allowances, deposits.
5. Finance page — principal, APR, term, payment schedule, fees, lender reference.
6. Scope page — design, permits, roof, electrical, trenching, inspections, utility work.
7. Warranty page — separate product, performance, workmanship, labor, roof and service terms.
8. Fine print — contingencies, cancellation, change orders, transfer, exclusions, document priority.

If the seller uses a web portal, save the proposal and disclosures locally before the link expires or changes. Record the file date. A revised quote should receive a new version label; do not silently overwrite the original.

4. Build the extraction sheet

Your extraction sheet has three columns in substance even if you write it as a list: the field, the exact stated value, and the source reference. Add an evidence label when the status is not obvious.

Commercial fields

Capture transaction type, gross cash price, discounts, customer deposits, financed principal, APR, term, all payment levels, expected prepayments, lease or PPA rates and escalators, total payments if disclosed, cancellation terms, and transfer conditions.

Do not subtract a possible tax credit from the cash price on this sheet. Put incentives in a separate, clearly dated section. This preserves a comparable price even if eligibility, timing, or law changes.

System fields

Capture DC system size, AC inverter capacity when stated, module count and wattage, exact module and inverter models, optimizer or microinverter models, battery usable capacity when relevant, mounting type, monitoring, array locations, and whether substitutions are allowed.

“Premium panels” is a claim. “20 Helio H410 modules” is a defined specification. If the exact model is missing, ask for it before comparing warranties or performance.

Production fields

Capture Year-1 annual production, the period represented, usage baseline, offset definition, weather source, shade or site model, azimuth and tilt if shown, system losses, inverter clipping treatment, degradation, downtime or availability assumption, export-credit assumption, utility escalation, and any guarantee with remedy.

Many proposals will not expose every modeling input. Do not invent missing values. Mark them MISSING and ask for the production report or assumption page.

Scope fields

Capture roof inspection, roof repair, removal and reinstallation, structural work, main-panel or service upgrade, trenching, conduit routing, attic work, permits, inspections, interconnection, utility fees, homeowner-association submissions, monitoring setup, commissioning, cleanup, and responsibility for failed inspections or redesign.

The purpose is not to demand that every item be included. It is to make price exposure visible.

Warranty and service fields

Capture warranty category, covered component or work, term, start date, responsible party, remedy, labor, shipping, diagnosis, removal and reinstallation, roof penetrations, transfer, registration requirements, exclusions, and service-response commitment if any.

Do not merge manufacturer and installer obligations. If a module maker supplies a replacement part but the installer does not cover labor, “warranty coverage” can still leave a homeowner with a bill.

Missing is a productive answer

The extraction sheet is successful when it reveals uncertainty early. A blank field is not a personal failure. It is a generated question. Write “not located” rather than guessing.

> EXTRACTION STANDARD
> Copy first, interpret second. Preserve units. Preserve whether a price is cash, financed, gross, or “net.” Preserve whether capacity is DC or AC. Preserve whether production is Year 1, average annual, or lifetime. Preserve whether a payment is initial, fixed, or conditional.

PART III — NORMALIZE PRICE AND FINANCING

5. Cash price is the common anchor

For purchase proposals, ask every installer for the cash price of the same defined scope. That number gives you a common commercial anchor even if you intend to finance.

Cash price does not mean “best price,” and cost per watt does not make scope differences disappear. It simply prevents a monthly-payment headline from replacing the underlying installation price.

Cash cost per watt

Formula:

CASH COST PER WATT = CASH PRICE / (DC SYSTEM SIZE IN kW x 1,000)

For the synthetic proposal:

$24,600 / (8.2 x 1,000) = $3.00 per watt.

DOE consumer guidance describes this normalization method when comparing installer quotes. Use it as a starting point, not a ranking. It becomes misleading when proposals bundle materially different items such as batteries, roofing, service upgrades, trenching, or unusually broad warranties.

Before comparing, annotate unequal scope. If one quote includes a $2,000 electrical allowance, you may show both the proposal's raw cost per watt and an adjusted comparison view—but never alter the original price field.

What not to put in the numerator

Do not automatically use:

- a “net” price after a presumed tax credit;
- financed principal when comparing installation cash prices;
- lifetime scheduled payments;
- price after a rebate that has uncertain eligibility;
- a combined solar-plus-battery price against a solar-only quote.

Each of those may be useful elsewhere. None is the same as cash installation price for equal scope.

6. Separate principal, payment, and total obligation

Three financing numbers answer three different questions.

PRINCIPAL or AMOUNT FINANCED asks: How much debt begins?

PAYMENT SCHEDULE asks: What must be paid and when, and can that amount change?

TOTAL OF PAYMENTS or the controlling equivalent asks: What do the scheduled credit payments add up to under the disclosed terms?

A low payment can coexist with a high principal, long term, expected prepayment, or future payment change. Do not infer one from another.

Financed premium over cash

Formula:

FINANCED PREMIUM = (FINANCED PRINCIPAL - CASH PRICE) / CASH PRICE x 100%

Synthetic example:

($31,980 - $24,600) / $24,600 = 30.0%.

This calculation says that the starting principal is 30% above the cash price. It does not identify the cause, calculate interest, establish illegality, or prove deception. The proper next action is a written request for a same-scope cash price and an explanation of the difference.

Simple scheduled-payment arithmetic

If a disclosure shows one fixed payment for a fixed number of months, multiplying the two is a useful check:

$211 x 240 = $50,640.

Label this “simple scheduled total,” not “true lifetime cost.” It can be wrong or incomplete when payments change, a balloon amount exists, fees sit outside the payment stream, an expected prepayment changes amortization, or the official disclosure uses a different schedule. The lender's actual disclosure controls.

The expected-prepayment test

Ask four direct questions:

- Is a large voluntary or expected prepayment assumed?
- By what date?
- What happens to the required payment if I do not make it?
- Where is that change shown in the lender's documents?

Do not assume a future tax benefit will arrive as a cash refund available for prepayment. Personal eligibility and tax liability are outside this book and must be checked independently.

APR is necessary, not sufficient for proposal extraction

APR is an important standardized credit measure, but a solar comparison still needs the cash price, principal, payment schedule, and total-of-payments disclosure. A low stated rate does not by itself explain why a financed principal differs from the cash price. Likewise, a higher rate does not automatically make an offer worse if the principal and fees differ.

This book does not calculate an optimal loan. It makes the disclosed structures comparable enough for you—or an independent adviser—to evaluate them.

7. Ownership, transfer, and exit

Solar is attached to a home, while the financing or service arrangement may last many years. Therefore the question “What if I sell?” belongs in the first comparison, not in a future folder.

For an owned system with a loan, locate payoff and any lender security-interest terms. Ask whether assumption by a buyer is permitted and what approvals are required. For a lease or PPA, locate transfer qualification, fees, timelines, buyout options, early termination, and end-of-term removal.

Do not accept “easy transfer” without a process. Convert it into:

“Please identify the contract section governing a home sale, the documents and buyer qualifications required, all fees, the typical processing time, and the alternative if transfer is not approved.”

Also identify who owns renewable-energy certificates, tax benefits, rebates, or other attributes. Do not assume the homeowner receives them merely because the system is on the roof.

PART IV — CLAIM DECODER

8. How to read a sales claim without overreacting

Marketing language compresses. A short phrase can refer to several possible structures. The disciplined response is neither trust nor accusation. It is definition.

Use the same four-step decoder every time:

CLAIM -> WHAT IT MAY MEAN -> WHAT TO VERIFY -> QUESTION TO ASK

“Low monthly payment”

WHAT IT MAY MEAN

A long term; a temporary payment level; an expected prepayment; a larger financed principal paired with a lower stated rate; or simply a competitive fixed-payment loan.

WHAT TO VERIFY

Cash price for identical scope; financed principal; APR; term; every payment level and date; fees; expected prepayment; re-amortization; balloon payment; official total of payments; prepayment terms.

QUESTION TO ASK

“Please provide the lender disclosure showing principal, APR, all payment amounts and dates, total of payments, fees, and every event that changes the required payment.”

“No money down”

WHAT IT MAY MEAN

No payment at installation. A loan, lease, PPA, lien, long-term payment obligation, or cancellation exposure may still begin.

WHAT TO VERIFY

Transaction type; ownership; contract length; principal or payment rate; escalation; security interest; cancellation; transfer; payoff; end-of-term obligations.

QUESTION TO ASK

“What financial or contractual obligation begins when I sign, and what must be paid, transferred, purchased, or removed if I sell the home or end the agreement?”

“25-year warranty”

WHAT IT MAY MEAN

A module performance warranty; module product warranty; inverter warranty; installer workmanship warranty; roof-penetration warranty; labor coverage; or several separate terms with different providers.

WHAT TO VERIFY

Covered item; triggering failure; term; start date; exclusions; claim procedure; remedy; parts; diagnosis; shipping; labor; removal and reinstallation; responsible party; transfer and registration.

QUESTION TO ASK

“Please separate every warranty by covered item, term, exclusions, remedy, labor and shipping coverage, and the company responsible for honoring it.”

“100% offset”

WHAT IT MAY MEAN

A modeled annual ratio in which estimated solar production equals a selected annual-usage baseline. The definition may use recent usage, adjusted future usage, or another period.

WHAT TO VERIFY

Usage baseline; annual versus monthly definition; Year-1 production; weather and shade inputs; system losses; degradation; future load changes; utility fixed charges; export-credit assumptions; guarantee and remedy.

QUESTION TO ASK

“Please define offset, show the usage baseline and period, and list every production and utility assumption used to calculate the percentage.”

“Turnkey”

WHAT IT MAY MEAN

The installer coordinates the normal project from design through permission to operate. The label may still exclude roof repairs, structural work, electrical upgrades, trenching, utility charges, homeowner-association work, hazardous materials, or correction of undisclosed conditions.

WHAT TO VERIFY

Included and excluded scope; allowances; subcontractors; permits; inspections; interconnection; utility and HOA tasks; redesign; change orders; cleanup; monitoring activation; responsibility after failed inspection.

QUESTION TO ASK

“Please replace ‘turnkey’ with a written included/excluded scope and identify every condition that can change the price or homeowner responsibility.”

“Free battery” or “included at no cost”

WHAT IT MAY MEAN

A discount, bundle, manufacturer promotion, incentive assumption, financed cost, or marketing allocation in which the battery has no separately displayed price.

WHAT TO VERIFY

Cash price with and without the battery; financed principal with and without it; model and usable capacity; installation scope; backup loads; warranty; incentive conditions; ownership; removal or replacement obligations.

QUESTION TO ASK

“Please show the same-scope cash and financed prices with and without the battery and identify the promotion or incentive conditions supporting the ‘free’ description.”

“Eliminate your electric bill”

WHAT IT MAY MEAN

A modeled claim that reduced grid purchases or credits may offset much of an energy charge under selected assumptions.

WHAT TO VERIFY

Utility fixed charges; tariff; export compensation; seasonal production and consumption; system size; load changes; escalation assumptions; financing payment; guarantee.

QUESTION TO ASK

“Please identify every utility charge expected to remain and separate estimated utility savings from loan, lease, or PPA payments.”

“Locked-in energy rate”

WHAT IT MAY MEAN

A fixed PPA rate, a starting rate with an escalator, or a comparison against an assumed future utility rate.

WHAT TO VERIFY

Starting rate; escalation percentage and timing; contract length; minimum charges; production basis; utility charges that remain; renewal and termination.

QUESTION TO ASK

“Please provide the complete payment-rate schedule for every contract year and the assumptions used for the utility-rate comparison.”

“Tier 1,” “premium,” or “best-in-class” equipment

WHAT IT MAY MEAN

A seller-defined quality statement, a reference to a third-party classification with a particular purpose, or simply a marketing category.

WHAT TO VERIFY

Manufacturer and model; datasheet; product and performance warranty; efficiency; degradation terms; temperature coefficient if relevant; authorized substitutions; service process.

QUESTION TO ASK

“Please provide exact model numbers and datasheets and explain which measurable specification supports the quality label used in the proposal.”

“Production guarantee”

WHAT IT MAY MEAN

A contractual minimum, a limited reimbursement formula, a monitoring commitment, or a sales estimate incorrectly described in conversation as a guarantee.

WHAT TO VERIFY

Guaranteed quantity and period; measurement source; exclusions; degradation schedule; claim window; homeowner maintenance duties; remedy; cap; responsible party.

QUESTION TO ASK

“Please provide the controlling guarantee terms, including the measured threshold, exclusions, claim process, remedy, and maximum payment.”

“Price valid today”

WHAT IT MAY MEAN

A real expiration, equipment allocation, promotion, financing rate lock, or sales-pressure device. The phrase alone does not distinguish among them.

WHAT TO VERIFY

Written expiration; what changes afterward; cancellation rights; deposit refundability; site-survey contingencies; financing approval; equipment availability.

QUESTION TO ASK

“Please identify which specific price or term expires, the reason and date, and all cancellation and refund terms if site findings or financing change the offer.”

> DECODER DISCIPLINE
> Never write “scam” in your comparison merely because a phrase is broad. Write the missing definition and request the controlling document. If a seller refuses to clarify a material term in writing, the refusal—not your speculation—is decision evidence.

PART V — SYSTEM AND PRODUCTION

9. Identify the system you are actually comparing

An equipment comparison begins with exact identities, not brand impressions.

For modules, capture manufacturer, model, quantity, watts per module, total DC capacity, product-warranty term, performance-warranty term, and any permitted substitute. Confirm that quantity multiplied by module watts matches the stated DC size, allowing for ordinary rounding.

For inverters, capture architecture—string inverter, microinverters, or another configuration—plus manufacturer, model, AC rating, warranty, monitoring, optimizer details if applicable, and replacement labor responsibility. Do not infer that a larger DC-to-AC ratio is good or bad without design context; use it as a question when proposals differ materially.

For batteries, capture manufacturer and model, usable energy capacity, continuous and surge power when stated, backup configuration, protected loads, operating mode, warranty throughput or cycles where applicable, installation location, thermal or ventilation requirements, and whether the battery can operate during an outage. “Battery included” does not define backup capability.

For mounting and electrical work, capture attachment system, roof areas used, conduit route, service-panel work, disconnects, meter work, trenching, structural review, and any reroofing coordination.

The substitution clause

Supply changes happen. The key question is not whether substitution is imaginable; it is who may approve it and what “equivalent” means.

Ask:

“Does the contract permit equipment substitutions? If so, please define the minimum technical and warranty specifications and confirm that I approve any material substitution in writing before installation.”

Do not compare Model X in a proposal if the contract lets the installer substitute an undefined equivalent without your approval.

Compare functions before specifications

Some equipment differences matter because they change the system's function: outage behavior, module-level monitoring, expansion, roof layout, service process, or battery capability. Other differences may be technically real but not decision-critical for your home.

Write the buyer question first: “Which loads must operate in an outage?” or “Who replaces a failed inverter and who pays labor?” Then compare the specification that answers it.

This prevents the worksheet from becoming a catalog of numbers with no decision purpose.

10. Decode the production estimate

An annual production estimate is a model output. It is useful, but it is not direct evidence of future utility savings and is not automatically a guarantee.

NREL's PVWatts tool, for example, estimates energy production for grid-connected photovoltaic systems and exposes inputs and losses. Installer models may use different tools and site-specific data. Two proposals can therefore produce different estimates even when their hardware is similar.

Production per installed kW

Formula:

YEAR-1 PRODUCTION INTENSITY = ESTIMATED YEAR-1 kWh / DC SYSTEM SIZE IN kW

Synthetic example:

11,480 kWh / 8.2 kW = 1,400 kWh per installed kW.

This metric highlights differing assumptions or designs. It does not measure equipment quality, guarantee output, or establish savings. A high result may reflect favorable orientation, low shade, a strong design, or optimistic inputs.

The assumption stack

Request or locate:

- address and array geometry;
- roof plane, orientation or azimuth, and tilt;
- shade study or remote shade assumption;
- weather dataset;
- module and inverter models;
- DC and AC capacity;
- soiling, shading, snow, mismatch, wiring, availability, and other system losses;
- inverter clipping treatment;
- Year-1 definition;
- annual degradation;
- downtime or availability;
- future tree growth or roof obstructions;
- production guarantee, if any.

NREL documentation explains that modeled system losses can include soiling, shading, snow, mismatch, wiring, connections, nameplate effects, system age, and operational availability. You do not need to become a modeling engineer. You need enough disclosure to understand why two estimates differ.

Offset is not a bill forecast

An annual offset percentage usually compares modeled production with a chosen annual consumption baseline. It does not by itself show monthly matching, export compensation, fixed utility charges, financing payments, future load growth, outages, or tariff changes.

The FTC advises consumers to examine utility arrangements and notes that fixed utility charges may remain even when solar reduces purchased kilowatt-hours. Therefore keep three rows separate:

- estimated energy production;
- assumed household consumption;
- modeled financial effect under a specific utility rule.

If a proposal jumps directly from kWh to dollars, ask for the tariff, export-credit, fixed-charge, escalation, and time-period assumptions.

Compare guarantees separately

A proposal estimate of 11,480 kWh is not the same as a contractual guarantee of 11,480 kWh. If a guarantee exists, extract its threshold, measurement period, degradation schedule, exclusions, claim window, required maintenance, remedy, cap, and responsible party.

Do not increase an estimate's status because the chart looks precise. Precision of presentation is not certainty of outcome.

PART VI — SCOPE AND WARRANTIES

11. Scope is where price differences become real

The most useful scope question is not “Is this turnkey?” It is “Which party is responsible for each task and each foreseeable exception?”

Roof

Record roof material, approximate age if known, observed condition, installer inspection, structural review, attachment method, flashing, leak responsibility, roof-penetration coverage, and any need to remove and reinstall panels for roof work.

DOE consumer guidance recommends asking whether roof repairs are needed and who is responsible if damage or a leak develops. Ask for roof-related work to be included in the proposal when applicable.

Pause if a roof appears near replacement but every proposal assumes installation without addressing reroofing. This is not a diagnosis; it is a coordination gap requiring qualified assessment.

Electrical service

Record main-panel capacity, service upgrade, subpanel, meter or disconnect work, utility requirements, code corrections, grounding, trenching, and allowance limits. If the quote says “panel upgrade if required,” ask who decides, when, at what price, and whether you approve a change order before work.

Permits, inspections, and interconnection

Separate preparation from fees and separate submission from approval. Record who prepares drawings, files permits, pays fees, responds to corrections, schedules inspections, submits interconnection paperwork, resolves utility redesign, and obtains permission to operate.

"Interconnection included" may mean paperwork labor is included while utility upgrade costs are not. Ask both.

Site conditions and construction

Check trenching, attic access, conduit appearance, landscaping, tree work, hazardous materials, structural reinforcement, equipment location, internet requirements, cleanup, damage repair, and work hours. Capture any allowance and the change-order method.

Project schedule

Record milestones rather than one promised completion date: site survey, final design, permit submission, financing approval, equipment availability, installation, inspection, utility approval, monitoring activation, and permission to operate.

A seller cannot control every authority's timing. The useful comparison is which dependencies are disclosed, which party owns each action, and what happens after delay.

12. Build a warranty responsibility map

Warranty comparison fails when all terms are reduced to one number. Build a separate row for each category.

Product warranty

This generally concerns defects in a component. Identify the manufacturer, product, term, exclusions, claim process, replacement remedy, shipping, labor, diagnosis, and removal/reinstallation.

Performance warranty

This generally concerns a module maker's warranted power-output level over time. It is not necessarily a guarantee of your system's annual energy production. Ask what is measured, by whom, and what remedy applies.

Workmanship warranty

This concerns the installer's work. Identify the legal company, term, start date, covered defects, exclusions, roof penetrations, service response, transfer, and remedy. A long manufacturer warranty does not extend a short workmanship warranty.

Roof or penetration coverage

Do not assume this is inside workmanship. Ask whether leaks, flashing, removal and reinstallation, interior damage, and coordination with a roofer are covered, for how long, and by whom.

Production guarantee

Keep this outside the equipment-warranty rows. Extract the promised production threshold, degradation, measurement system, exclusions, homeowner duties, remedy, and cap.

Service and labor

Ask who diagnoses a problem, who opens the manufacturer claim, who supplies labor, who pays truck-roll or shipping costs, and what happens if the original installer no longer operates.

The DOE emphasizes installer credentials, experience, transparency, subcontractor oversight, roof responsibility, and warranty explanation. These are not a substitute for local license verification or contract review; they are prompts for your evidence packet.

> WARRANTY TEST
> For every 20- or 25-year headline, complete this sentence: "If ___ happens in year ___, ___ company must provide ___, while I may still pay ___." If you cannot complete it from the documents, the headline is not yet comparable.

PART VII — THE CALCULATION LAYER

13. Four metrics, four limited jobs

Calculations are useful only when their limits remain attached.

Metric 1: cash cost per watt

Question answered: How does the quoted cash installation price relate to DC system size?

Formula: cash price / DC watts.

Useful when: scope is comparable and storage or unrelated construction is separated.

Limit: does not value equipment, production quality, roof work, financing, service, or warranty. A lower number is not automatically a better offer.

Metric 2: financed premium over cash

Question answered: How much larger is the starting financed principal than the same-scope cash price?

Formula: (principal - cash price) / cash price.

Useful when: comparing the commercial effect of financing structures.

Limit: does not calculate interest or explain the difference. A zero premium can accompany a higher APR; a premium can reflect fees or a changed scope. Verify.

Metric 3: scheduled payments

Question answered: What do the stated scheduled payments add up to under the disclosed schedule?

Formula: sum each payment amount multiplied by its number of periods, plus disclosed balloon or terminal payments.

Useful when: checking the order of magnitude and comparing payment paths.

Limit: use the lender's official disclosure. The simple sum does not model early payoff, taxes, insurance, opportunity cost, utility bills, or payments that are not fixed.

Metric 4: Year-1 production per installed kW

Question answered: How aggressive or productive is the Year-1 estimate relative to DC system size?

Formula: estimated Year-1 kWh / DC kW.

Useful when: proposals show different sizes and output estimates.

Limit: not a quality score, savings guarantee, or universal benchmark. Compare site and modeling assumptions.

Optional metric: price per estimated Year-1 kWh

Formula: cash price / estimated Year-1 kWh.

This can be a diagnostic view, but it combines a fixed price with one modeled year. It rewards optimistic production assumptions and ignores degradation, lifespan, financing, utility value, and scope. Use it only to generate questions—not to rank offers.

14. Calculation controls

Before calculating, run five controls:

1. Same unit? Convert every system to DC watts or DC kW.
2. Same scope? Separate batteries and major construction where possible.
3. Same time basis? Distinguish Year 1, average annual, and lifetime totals.
4. Same transaction field? Do not mix cash price, principal, and total payments.
5. Same assumption status? Mark modeled values as estimates.

Round only after calculating. Two decimal places are generally enough for cost per watt; whole kWh is enough for annual production; one decimal point is enough for a percentage diagnostic. Do not create false precision.

If a number does not reconcile, do not repair the proposal silently. Example: 20 modules x 410 W = 8.2 kW. If the summary says 8.6 kW, record the mismatch and ask which equipment schedule is correct.

PART VIII — WORKED THREE-QUOTE DECISION

15. The synthetic case

Jordan has three rooftop-solar purchase proposals. All names and values are invented. The offers are intentionally non-obvious.

PROPOSAL A — LOWEST CASH PRICE

Cash price $24,600. System 8.2 kW DC. Year-1 estimate 11,480 kWh. Financed principal $31,980. Stated payment $211 for 240 months if fixed. Standard permit and installation scope. Main-panel upgrade excluded. Workmanship warranty 10 years.

PROPOSAL B — BROADER ELECTRICAL ALLOWANCE

Cash price $26,400. System 8.0 kW DC. Year-1 estimate 10,960 kWh. Financed principal $26,400. Stated payment $293 for 144 months if fixed. Main-panel work included up to a $2,000 allowance. Workmanship warranty 15 years.

PROPOSAL C — LOWEST PAYMENT, HIGHEST OUTPUT INTENSITY

Cash price $25,200. System 7.6 kW DC. Year-1 estimate 11,400 kWh. Financed principal $28,980. Stated payment $169 for 300 months if fixed. Roof work excluded. Workmanship warranty 5 years. Production model uses a remote shade assumption not attached to the proposal.

First-pass economics

A cash cost per watt: $24,600 / 8,200 = $3.00/W.

B cash cost per watt: $26,400 / 8,000 = $3.30/W.

C cash cost per watt: $25,200 / 7,600 = $3.32/W.

A financed premium: ($31,980 - $24,600) / $24,600 = 30.0%.

B financed premium: ($26,400 - $26,400) / $26,400 = 0.0%.

C financed premium: ($28,980 - $25,200) / $25,200 = 15.0%.

Simple scheduled totals, only if every stated payment is fixed:

A: $211 x 240 = $50,640.

B: $293 x 144 = $42,192.

C: $169 x 300 = $50,700.

These results do not select B. B has a higher cash price and a different electrical allowance. The loan APRs and official disclosures have not been compared. The simple totals could omit a payment change or other term.

Production normalization

A: 11,480 / 8.2 = 1,400 kWh per installed kW.

B: 10,960 / 8.0 = 1,370 kWh per installed kW.

C: 11,400 / 7.6 = 1,500 kWh per installed kW.

C's higher intensity is a reason to inspect its model, not a reason to reward or reject it. The missing shade report matters. B's lower figure may be conservative, may reflect greater shade, or may reflect a different layout. The proposals have not supplied enough evidence to know.

Scope normalization

A excludes a possible main-panel upgrade. B includes an allowance but does not state what happens above $2,000. C excludes roof work and has not documented roof condition. Therefore the raw cash prices do not represent equal risk.

The correct output is three questions:

- A: “Please state the fixed price or pricing method for any required main-panel upgrade and when that need will be determined.”
- B: “Please define what the $2,000 allowance covers and how any overage is approved and priced.”
- C: “Please state the roof-condition assumption, who determines whether repair is required, and the price and schedule effect if work is needed.”

Warranty normalization

A and B identify separate workmanship terms. C leads with a 25-year warranty elsewhere but lists only five years of workmanship. That is not necessarily contradictory: the 25-year term may concern panel performance. It must be separated.

Question to C:

“Please separate the 25-year term into product, performance, workmanship, labor, and roof-penetration coverage, with the responsible company and exclusions for each.”

The decision after replies

Assume the written replies arrive:

- A estimates a $3,100 main-panel upgrade if required and confirms the loan payment rises in month 19 unless Jordan makes a stated prepayment.
- B confirms the panel allowance and quotes a written cap for the likely work; its loan disclosure shows one fixed payment schedule.
- C supplies a shade model that assumes removal of a tree not included in scope and confirms only five years of workmanship labor.

The packet now contains decision evidence. Jordan may value B's broader scope and simpler payment schedule, A's lower cash price if paying cash and the panel upgrade proves unnecessary, or C's smaller system if the tree work and assumptions are resolved. This book does not decide among those preferences.

The important result is that the choice is no longer between three headlines. It is between three documented packages.

PART IX — QUESTION GENERATOR

16. Turn gaps into answerable questions

Weak question: “Is everything included?”

Strong question: “For Proposal B dated October 3, please list every permit, inspection, interconnection, utility, electrical, roof, trenching, and monitoring item included in the cash price, and identify every condition that can create a change order.”

A strong written question contains four elements:

1. Proposal identity and date.
2. Exact field or conflict.
3. Requested definition, number, document, or responsibility.
4. Request to identify any effect on price, schedule, design, or contract.

Price and financing questions

- “Please provide the gross cash price before customer-specific incentives and the financed principal for identical scope.”
- “Please explain in writing every difference between the cash price and amount financed.”
- “Please provide the lender disclosure showing APR, term, every payment level, total of payments, fees, expected prepayment, and any re-amortization or balloon event.”
- “Does early payoff change any fee, rate, or obligation?”
- “Which document controls if the proposal payment conflicts with the lender schedule?”

Production questions

- “Please provide the site or shade report used for the Year-1 estimate.”
- “Which weather dataset, system-loss assumption, degradation rate, and availability assumption were used?”
- “How is offset defined, and which consumption period supplies the baseline?”
- “Which fixed charges and export-credit rules remain in the utility-bill estimate?”
- “Is any production level contractually guaranteed? If yes, provide the threshold, exclusions, remedy, and cap.”

Scope questions

- “When will roof and electrical conditions be verified, and can I review resulting design or price changes before work?”
- “Who pays for permits, corrections, inspection rework, interconnection fees, utility upgrades, trenching, and HOA requirements?”
- “What is expressly excluded?”
- “Which subcontractors may perform work, and who remains responsible for them?”
- “What event allows a change order, how is price determined, and must I approve it in writing?”

Warranty questions

- “Please separate product, performance, workmanship, labor, roof-penetration, and production coverage.”
- “Who diagnoses a failure and opens a manufacturer claim?”
- “Who pays labor, shipping, access, removal, and reinstallation?”
- “What registration, maintenance, monitoring, or notice is required to keep coverage?”
- “What happens to service obligations if the original installer is unavailable?”

Transfer and cancellation questions

- “What can I cancel, by when, using which method, and what deposit or fee is refundable?”
- “If I sell the home, what must be paid off, assumed, transferred, purchased, or removed?”
- “What buyer qualification, processing time, fee, and alternative applies if transfer is not approved?”

17. Manage replies as proposal revisions

An email answer is useful evidence, but it may not amend the contract. When a response changes a material term, request a revised proposal, addendum, lender disclosure, or contract section.

Use this reply sequence:

1. Log date sent and date received.
2. Attach the reply to the relevant proposal.
3. Mark each question ANSWERED, PARTIAL, CONFLICT, or UNANSWERED.
4. Copy the answer into the comparison only with its source reference.
5. Request a revised controlling document for material changes.
6. Recalculate only after the revised numbers are documented.

Do not let version 3 erase version 1. A version history shows what changed and can reveal whether a lower price came from narrower scope or different equipment.

A sendable request

Subject: Written clarifications for Proposal [ID and date]

Thank you for the proposal. I am comparing the written price, financing, production assumptions, scope, and warranty terms across several offers. Please answer the questions below in writing. For any answer that changes the quoted price, equipment, production estimate, schedule, or responsibility, please provide a revised proposal or addendum.

[Insert numbered questions from your question sheet.]

Please also identify which installation contract, lender disclosure, lease, PPA, warranty, or addendum will control each answer.

Thank you.

PART X — YOUR DECISION PACKET

18. Packet instructions

The following forms support one comparison of up to three proposals. They are intentionally not duplicated into multiple giant packets. If you need another comparison, photocopy the blank pages for personal use before writing, or reproduce the same field structure in your own notes.

Complete the packet in this order:

1. Proposal register and document inventory.
2. Extraction pages for A, B, and C.
3. Normalized price and financing comparison.
4. System and production comparison.
5. Scope and warranty comparison.
6. Question sheets for each seller.
7. Answer and revision log.
8. Unresolved-items and pause-condition review.
9. Decision record.

Do not fill every box merely because it exists. Write N/A when a field truly does not apply, NOT LOCATED when it may apply but is absent, and VERIFY when the proposal's meaning is unclear.

Proposal register

Write the exact stated value and its proposal page/section. Use NOT LOCATED rather than guessing.

Property / project	
Proposal A — company, ID, date	
Proposal B — company, ID, date	
Proposal C — company, ID, date	
Decision deadline claimed	
My actual decision deadline	
Files saved at / folder	
People included in review	

Document inventory

Mark RECEIVED, REQUESTED, N/A, or NOT LOCATED for each proposal.

Documents supplied

FIELD	PROPOSAL A	PROPOSAL B	PROPOSAL C
Full proposal + version date			
Installation contract			
Cash-price page			
Lender disclosure / agreement			
Lease or PPA agreement			
System layout			
Equipment datasheets			
Production / shade report			
Warranty documents			
Scope + exclusions			
Cancellation / transfer terms			

Proposal A: Commercial extraction

Write the exact stated value and its proposal page/section. Use NOT LOCATED rather than guessing.

Transaction type	
Gross cash price	
Discounts / adders	
Financed principal	
APR and term	
Payment schedule	
Expected prepayment / change	
Lease/PPA rate + escalator	
Cancellation / transfer	

Proposal A: System and production extraction

Write the exact stated value and its proposal page/section. Use NOT LOCATED rather than guessing.

DC system size	
Module make/model/count	
Inverter make/model/AC size	
Battery model/capability	
Year-1 production	
Usage baseline / offset	
Shade/weather source	
Loss + degradation assumptions	
Guarantee + remedy	

Proposal A: Scope and warranty extraction

Write the exact stated value and its proposal page/section. Use NOT LOCATED rather than guessing.

Roof / structural	
Electrical / service	
Permits / inspections	
Interconnection / utility	
Trenching / site work	
Allowances / change orders	
Product/performance warranties	
Workmanship / roof coverage	
Labor / responsible party	

Proposal B: Commercial extraction

Write the exact stated value and its proposal page/section. Use NOT LOCATED rather than guessing.

Transaction type	
Gross cash price	
Discounts / adders	
Financed principal	
APR and term	
Payment schedule	
Expected prepayment / change	
Lease/PPA rate + escalator	
Cancellation / transfer	

Proposal B: System and production extraction

Write the exact stated value and its proposal page/section. Use NOT LOCATED rather than guessing.

DC system size	
Module make/model/count	
Inverter make/model/AC size	
Battery model/capability	
Year-1 production	
Usage baseline / offset	
Shade/weather source	
Loss + degradation assumptions	
Guarantee + remedy	

Proposal B: Scope and warranty extraction

Write the exact stated value and its proposal page/section. Use NOT LOCATED rather than guessing.

Roof / structural	
Electrical / service	
Permits / inspections	
Interconnection / utility	
Trenching / site work	
Allowances / change orders	
Product/performance warranties	
Workmanship / roof coverage	
Labor / responsible party	

Proposal C: Commercial extraction

Write the exact stated value and its proposal page/section. Use NOT LOCATED rather than guessing.

Transaction type	
Gross cash price	
Discounts / adders	
Financed principal	
APR and term	
Payment schedule	
Expected prepayment / change	
Lease/PPA rate + escalator	
Cancellation / transfer	

Proposal C: System and production extraction

Write the exact stated value and its proposal page/section. Use NOT LOCATED rather than guessing.

DC system size	
Module make/model/count	
Inverter make/model/AC size	
Battery model/capability	
Year-1 production	
Usage baseline / offset	
Shade/weather source	
Loss + degradation assumptions	
Guarantee + remedy	

Proposal C: Scope and warranty extraction

Write the exact stated value and its proposal page/section. Use NOT LOCATED rather than guessing.

Roof / structural	
Electrical / service	
Permits / inspections	
Interconnection / utility	
Trenching / site work	
Allowances / change orders	
Product/performance warranties	
Workmanship / roof coverage	
Labor / responsible party	

Normalized price comparison

FIELD	PROPOSAL A	PROPOSAL B	PROPOSAL C
Gross cash price			
DC system watts			
Raw cash $/W			
Battery / unequal scope			
Adjusted view (labeled)			
Deposit / milestones			
Largest price contingency			
Source references			

Financing / contract comparison

FIELD	PROPOSAL A	PROPOSAL B	PROPOSAL C
Amount financed / principal			
Difference vs cash			
APR			
Term			
Payment levels + dates			
Expected prepayment			
Official total of payments			
Fees / balloon			
Security interest			
Early payoff			
Sale / transfer			
Source references			

System comparison

FIELD	PROPOSAL A	PROPOSAL B	PROPOSAL C
DC / AC capacity			
Modules			
Inverter / optimizers			
Battery / backup loads			
Mounting / layout			
Monitoring			
Substitution clause			
Function that matters			
Source references			

Production-model comparison

FIELD	PROPOSAL A	PROPOSAL B	PROPOSAL C
Year-1 kWh			
kWh per installed kW			
Usage baseline			
Offset definition			
Shade / site source			
Weather source			
Loss assumptions			
Degradation / availability			
Utility/export assumptions			
Guarantee + remedy			
Unresolved difference			

Scope and responsibility

FIELD	PROPOSAL A	PROPOSAL B	PROPOSAL C
Roof / reroofing			
Structural work			
Main panel / service			
Trenching / conduit			
Permits / corrections			
Inspection / rework			
Interconnection / utility			
HOA / other approvals			
Monitoring / commissioning			
Exclusions / allowances			
Change-order control			

Warranty responsibility map

FIELD	PROPOSAL A	PROPOSAL B	PROPOSAL C
Module product			
Module performance			
Inverter / optimizer			
Battery			
Installer workmanship			
Roof penetrations			
Production guarantee			
Labor / diagnosis			
Shipping / removal			
Transfer / registration			
Responsible company			

Proposal A: written questions

Name the proposal and source. Ask for a number, definition, responsibility, or controlling document.

1. GAP / SOURCE	
QUESTION / DOCUMENT REQUEST	
2. GAP / SOURCE	
QUESTION / DOCUMENT REQUEST	
3. GAP / SOURCE	
QUESTION / DOCUMENT REQUEST	
4. GAP / SOURCE	
QUESTION / DOCUMENT REQUEST	
5. GAP / SOURCE	
QUESTION / DOCUMENT REQUEST	
6. GAP / SOURCE	
QUESTION / DOCUMENT REQUEST	

Proposal B: written questions

Name the proposal and source. Ask for a number, definition, responsibility, or controlling document.

1. GAP / SOURCE	
QUESTION / DOCUMENT REQUEST	
2. GAP / SOURCE	
QUESTION / DOCUMENT REQUEST	
3. GAP / SOURCE	
QUESTION / DOCUMENT REQUEST	
4. GAP / SOURCE	
QUESTION / DOCUMENT REQUEST	
5. GAP / SOURCE	
QUESTION / DOCUMENT REQUEST	
6. GAP / SOURCE	
QUESTION / DOCUMENT REQUEST	

Proposal C: written questions

Name the proposal and source. Ask for a number, definition, responsibility, or controlling document.

1. GAP / SOURCE	
QUESTION / DOCUMENT REQUEST	
2. GAP / SOURCE	
QUESTION / DOCUMENT REQUEST	
3. GAP / SOURCE	
QUESTION / DOCUMENT REQUEST	
4. GAP / SOURCE	
QUESTION / DOCUMENT REQUEST	
5. GAP / SOURCE	
QUESTION / DOCUMENT REQUEST	
6. GAP / SOURCE	
QUESTION / DOCUMENT REQUEST	

Answer and revision log

FIELD	PROPOSAL A	PROPOSAL B	PROPOSAL C
Question ID			
Sent / received			
Answer status			
Written answer source			
Price effect			
Scope/design effect			
Revised document + date			
Still unresolved			

Pause-condition review

A NO does not automatically reject the proposal. It means the decision packet is incomplete.

CHECK	YES / NO	EVIDENCE OR REQUIRED ACTION
Transaction / owner clear		
Cash vs financed basis clear		
All payment changes clear		
Incentive assumptions separated		
Exact system identified		
Production differences explained		
Roof/electrical exposure clear		
Scope/change orders clear		
Warranty responsibility clear		
Transfer/cancellation clear		
Conflicts resolved in writing		

Decision record

Record the reason, not just the winner. Acknowledge remaining uncertainty.

Decision date	
Selected proposal / no decision	
Transaction type	
Main reasons	
Trade-offs accepted	
Material facts relied on	
Documents controlling decision	
Remaining uncertainty	
Independent advice obtained	
Next action / deadline	

This record documents a process; it does not guarantee performance, savings, service, or legal outcome.

PART XI — FINAL REVIEW

19. Pause conditions before signing

Pause and resolve the issue if any of these conditions applies:

- You cannot identify the transaction type or system owner.
- The same-scope cash price is missing from a financed purchase comparison.
- The financed principal, payment schedule, or expected prepayment is unclear.
- A proposed tax benefit is treated as guaranteed personal cash.
- The system model or exact major equipment is missing or freely substitutable.
- The production estimate lacks enough assumptions to explain material differences.
- An offset or savings claim hides utility fixed charges, tariff, or export assumptions.
- Roof, electrical, trenching, utility, or change-order responsibility can materially change price but remains undefined.
- A long warranty headline has no clear covered event, responsible party, labor rule, or remedy.
- Home-sale transfer, payoff, cancellation, or end-of-term obligations are unresolved.
- A verbal assurance conflicts with the proposal, contract, or lender document.
- You are being asked to sign before receiving documents you requested.

Pausing does not mean rejecting solar or accusing a seller. It means the decision packet is not complete.

20. The final five-minute audit

Before choosing, ask:

1. Can I state what I am buying or agreeing to in one sentence?
2. Can I identify the gross cash price, financing obligation, or third-party payment schedule from a controlling document?
3. Can I explain why the production estimates differ without calling the highest number “best”?
4. Can I name the largest excluded or uncertain cost in each proposal?
5. Can I explain which company handles a component failure, workmanship problem, roof leak, and monitoring issue?
6. Have all material sales claims been converted into written definitions?
7. Are revised answers reflected in revised documents?
8. Do I understand cancellation, transfer, home-sale, and end-of-term obligations?

9. Have I separated confirmed facts from estimates, claims, and missing information?
10. Does my written reason for the decision acknowledge the remaining uncertainty?

If the answers are yes, the system has done its job. It has not guaranteed the outcome. It has converted a sales comparison into a buyer-owned record.

Glossary

AC — Alternating current. Inverter output is commonly expressed in AC terms.

APR — Annual percentage rate, a standardized measure associated with the cost of consumer credit. Read it with the principal, fees, payment schedule, and total-of-payments disclosure.

Azimuth — Compass direction an array faces, used in production modeling.

Cash price — The amount quoted for purchasing the defined installation without the proposal's financing structure and before customer-specific incentive assumptions.

Change order — A documented change to price, work, schedule, or design after the original agreement. Your contract determines its effect.

Clipping — Energy not converted when available DC power exceeds an inverter's usable AC output under particular conditions.

DC — Direct current. Solar-array nameplate capacity is commonly stated in DC watts or kilowatts.

Degradation — Modeled or warranted change in output capability over time.

Escalator — A contract term that increases a lease payment or PPA energy rate on a stated schedule.

Export compensation — The value or credit a utility provides for electricity sent to the grid. Rules vary and can change.

Financed principal — The starting amount financed, not the monthly payment and not the same as total scheduled payments.

Interconnection — The process and agreement for connecting a system to the utility grid.

kW — Kilowatt, a unit of power equal to 1,000 watts.

kWh — Kilowatt-hour, a unit of energy. Proposals often estimate annual kWh production.

Lease — A contract to use a system owned by another party, generally for recurring payments under stated terms.

Loss assumption — A modeled reduction associated with factors such as shading, soiling, wiring, mismatch, availability, or other system effects.

Offset — A proposal-defined relationship, often annual, between estimated solar production and an assumed electricity-use baseline.

Permission to operate — Utility authorization required before normal grid-connected operation under the applicable process.

PPA — Power purchase agreement, generally a contract to buy electricity produced by a system owned by another party.

Production estimate — Modeled energy output for a stated period and assumption set; not automatically a guarantee.

Re-amortization — Recalculation of scheduled loan payments based on contract terms and remaining balance, sometimes linked to an expected prepayment.

Scope — The work, materials, administrative tasks, assumptions, allowances, and exclusions assigned to each party.

System size — The proposal's stated capacity. Preserve whether it is DC or AC.

Workmanship warranty — Installer coverage for defined defects in installation work, distinct from manufacturer product or performance warranties.

Source and update note

This book's durable method was checked against current official consumer and technical sources on September 7, 2026. The book intentionally avoids depending on current incentive percentages, local export rates, utility programs, or advertised finance rates.

Primary sources:

1. U.S. Department of Energy, “Decisions, Decisions: Choosing a Solar Installer.” Guidance on installer qualifications, transparency, roof questions, proposal scope, comparing quotes, and cash cost per watt.
2. U.S. Department of Energy, “Homeowner's Guide to Solar” and consumer process resources. General system, financing, production-estimate, and consumer-document context. Because individual pages can retain older program language, current program facts must be checked with the responsible agency.
3. Consumer Financial Protection Bureau, “Issue Spotlight: Solar Financing,” August 7, 2024. Evidence on solar-specific loan presentation, financed-principal markups, tax-credit

assumptions, expected prepayments, re-amortization, and uncertain financial-benefit claims.

4. Federal Trade Commission, “Solar Power for Your Home.” Consumer questions covering utility charges, bids, system details, financing, leases, PPAs, warranties, repairs, transfer, and contract review.

5. Federal Trade Commission, “Solar energy is rising in popularity. So are the scams,” September 2024. Consumer warning against pressure, unsupported government or “free” claims, and signing before understanding terms.

6. National Renewable Energy Laboratory, PVWatts Calculator and technical documentation. Production-estimate purpose, model inputs, and loss categories.

7. U.S. Department of Energy, Weatherization Assistance Program Solar Bid Request Template, February 2024. A government procurement reference illustrating concrete fields such as system size, modules, inverters, production, price, warranty, and scope. This book does not reproduce that template.

Before relying on a proposal, refresh volatile information directly with the responsible source:

- IRS for federal tax rules and personal tax questions;
- your utility and public utility regulator for tariffs, fixed charges, export compensation, interconnection, and program status;
- state or local agencies for incentives, licensing, cancellation rules, and consumer protections;
- the lender or third-party owner for current financing, lease, or PPA terms;
- the manufacturer and installer for current warranty documents;
- qualified professionals for roof, electrical, engineering, tax, financial, insurance, or legal judgment.

No source can make a seller's proposal complete. The book's method remains: locate the statement, preserve its source, label its status, ask for the missing definition, and require material changes to appear in the controlling documents.